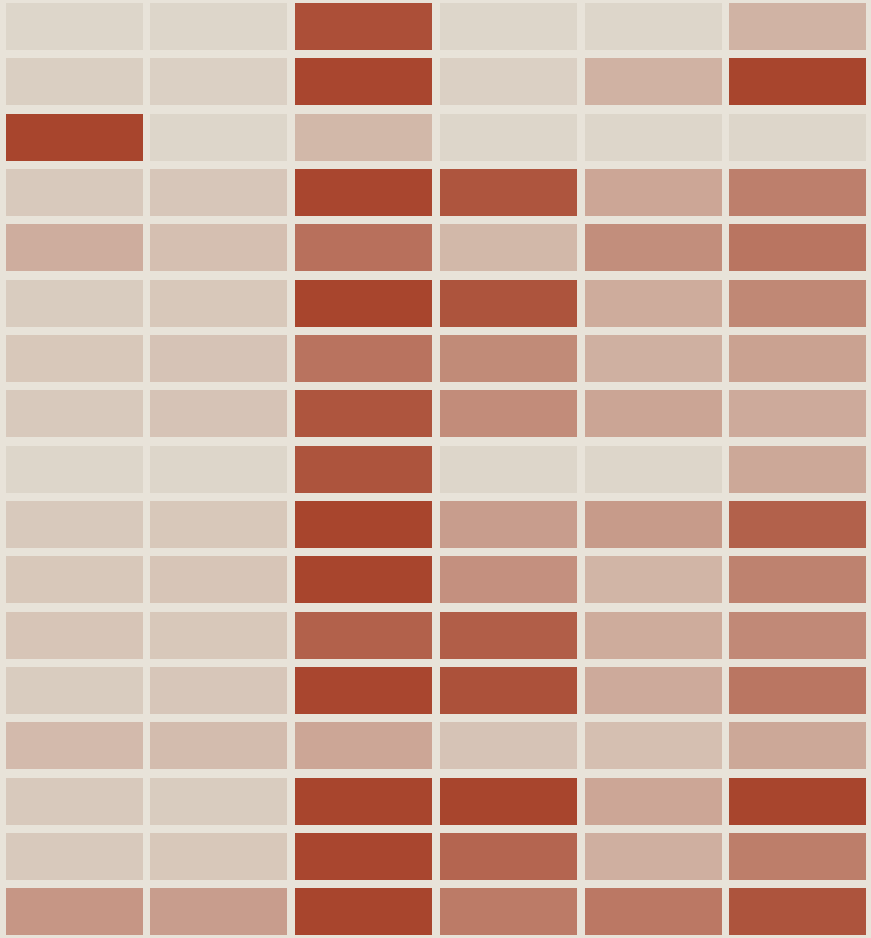
# Anomaly detection for data quality, measured inside the ETL pipeline

Living progress deck. Rule-based, classical and first deep detectors run through one pipeline and are compared per anomaly type, on a synthetic generator and on 42 real UCI tables with injected anomalies, and on 100 labelled problems of the pool.


| | |
|---|---|
| Author | Theo Bigand, 2026-11-11 |
| Repository | theobigand/etl-anomaly-benchmark |
| Evidence | pool of 152 open sets, synthetic generator, 17 detectors |

Mean recall, 17 detectors by 6 anomaly types, synthetic customer table, 5 seeds. Rows: Z-score to LUNAR as in the paper. Each family has a narrow band of dark cells.

# Does machine learning in an ETL pipeline beat plain rules, and for which anomalies?

The industry default is rule-based: auditable, cheap, and blind to what the author did not anticipate. Recent work adds ML detectors inside the flow, but the evidence is thin, one algorithm on one dataset. This thesis builds the missing head-to-head benchmark and breaks the result down by anomaly type.

What generic anomaly benchmarks leave out

**EVALUATED IN A VACUUM**

Algorithms scored outside any pipeline, no integration cost measured.

**NO DECLARATIVE BASELINE**

The rules that dominate production are never in the comparison.

**ONE AGGREGATE SCORE PER DATASET**

Hiding which family is good at which kind of defect.

**A SINGLE FLAG THRESHOLD**

Reported as one operating point, not the precision-recall trade-off.

# Benchmarks rank detectors. Few report their decisions

References were checked against a primary source or the DOI registry, with the limits listed in references.md. In the texts I read, ADBench and AnoShift report AUC only, and none of the works below breaks results down by data-quality fault. The scan is not systematic, so this is a gap I did not find filled, not a proof that nothing exists.

| Work | Time order | At a threshold |
|---|---|---|
| ADBench, 30 algorithms, 57 sets | not found | not found (AUC) |
| Agyemang 2024, one simulated set | none | precision, recall, F1 |
| AnoShift, Kyoto 2006-2015 | IID, NEAR, FAR | not found (AUC) |
| FedAD-Bench, federated deep | not found | F1-optimal threshold |
| Sun et al. 2024, online thresholds | shift in a stream | error-rate guarantees |

"Not found" means absent from the text I read, not proven absent. This thesis: six fault types, a controlled generator and real tables, decisions and pipeline cost.

4 RESEARCH QUESTIONS

# Four questions, one study each, hypotheses written first

Every study has a protocol and hypotheses committed before its runs. A hypothesis that fails is reported as failed: three of the six on the whole pool do.

| Question | Studies | Answer |
|---|---|---|
| RQ1 rules, classical, neural in one pipeline | examples, 100 labelled problems | historical ranking; train-only replay pending |
| RQ2 which anomaly type each family finds | synthetic matrix, schemas, 42 UCI tables | narrow competence, values do not transfer |
| RQ3 operating point and time | contamination, label-free threshold, chronological split | threshold fixed on the past fails when prevalence moves |
| RQ4 cost in the pipeline | cost study, ETL run on 152 sets | neural 10 to 100 times column-wise |

`The Results section follows these four questions; the Discussion answers them in one paragraph.`

# A controlled dataset that labels six anomaly types separately

12,360 records generated with Faker and NumPy, six types injected at 3% each, every row tagged with its category. Recall **and** precision are reported per type. The magnitudes are one **severity** multiplier, so the choice is swept rather than fixed.

**6**

anomaly types, labelled and measured separately

| | |
|---|---|
| **MISSING VALUE** | one or two feature cells set to null |
| **EXACT DUPLICATE** | a clean row copied, same business attributes |
| **UNIVARIATE OUTLIER** | one numeric field pushed far out of range |
| **MULTIVARIATE OUTLIER** | every field in range, the combination not (an 18-year-old with a top-percentile balance) |
| **TEMPORAL DRIFT** | recent-signup rows whose income and balance distribution has shifted up |
| **INVALID FORMAT** | malformed e-mail, out-of-domain country code, impossible age |

Because the category of every injected row is known, the benchmark reports a recall **and** a precision per type: the part that generic anomaly benchmarks, which publish one aggregate score, do not provide.

# One interface, one measured path

The demo and every benchmark call the same `detect()` stage, so latency is the real in-pipeline fit + predict + score time. A continuous `score()` on top of the boolean `predict()` makes ROC-AUC and AP computable; `detect(fit_df=...)` fits on the train slice and scores the test set. Exact deduplication belongs to the ordinary transform; typed injected duplicates instead remain in their separate evaluation with `transform_dedup`.

extract → transform → **detect** → load

```python
# every detector, one contract
class Detector(ABC):
    name: str
    def fit(self, df) -> Self: ...
    def predict(self, df) -> Series:   # bool per row
        ...
    def score(self, df) -> Series:     # graded score
        return self.predict(df).astype(float)

# classical ML: adapt PyOD, do not reimplement
class PyODDetector(Detector):
    def fit(self, df):
        x = impute(df[self.numeric_cols])
        self.model = self.estimator.fit(scale(x))
        return self

# registry: one line, picked up everywhere
"isolation_forest": lambda: IsolationForestDetector(),
```

6A METRIC INTERPRETATION

# Average Precision measures the ranking

AP uses continuous anomaly scores. Precision, recall, F1 and the flagged share use a selected decision threshold.

$$AP = \sum \Delta R \cdot P$$

precision weighted by each recall increment

### NON-INTERPOLATED

Distinct cutoffs run from high to low scores; tied scores enter together. AP differs from trapezoidal precision–recall area.

### PREVALENCE MATTERS

Prevalence is a no-skill reference precision. Constant scores give AP equal to prevalence; a random strict ranking has an upward finite-sample correction.

### SEPARATE DECISIONS FROM SCORES

Changing only the flag threshold leaves AP and ROC-AUC unchanged. ROC-AUC is complementary; neither ranking metric establishes a useful operating point.

### HISTORICAL NAMING

Stored `pr_auc` fields and remaining AP labels in archived figures and the paper denote this same AP quantity.

7 THREE FAMILIES, ELEVEN ROW-LEVEL DETECTORS

# Rules and classical models, one contract

Classical detectors use PyOD or the recorded protocol adaptations. The flag threshold uses training prevalence. They receive numeric columns only; Great Expectations separately checks nulls on all columns, so this is an explicit information asymmetry. ECOD, COPOD and HBOS aggregate marginal evidence without explicitly fitting dependence between columns. The time-aware `drift` detector is opt-in, synthetic only.

| Detector | Family | Principle | Main limit |
|---|---|---|---|
| Z-score | rule | flag `\|z\| > 3` on any column | mean and std inflated by outliers |
| IQR | rule | Tukey fences, `k = 1.5` | fires often in high dimension |
| Great Expectations | rule | null checks + fitted 0.1% / 99.9% bounds | generic statistical suite, no domain rules |
| Isolation Forest | joint | isolation depth under random splits | axis-aligned splits, weak on nulls |
| LOF | joint | local density vs 20 neighbours | cost grows with the scored set |
| One-Class SVM | joint | RBF boundary around the bulk | quadratic training |
| ECOD | column-wise | empirical tail probabilities, summed | no explicit cross-feature dependence |
| COPOD | column-wise | max of skew term and mean log tails | same |
| HBOS | column-wise | histogram per column, inverse bin height | independent columns, coarse bins |
| KNN | joint | distance to the 5th neighbour | quadratic, skipped above 50k rows |
| PCA | joint | reconstruction error, 95% of variance | misses a shift inside the kept space |
| drift | time-aware | recent-window mean shift | per row, so diluted drift is invisible |

7A FITTED RULES

# Column bounds do not fix row alert rates

All statistics come from training data. The evaluated Great Expectations suite is a generic statistical baseline, not a domain-informed rule set.

### Z-SCORE

Maximum |x − training mean| / sample standard deviation. Flag above 3; constant columns contribute zero.

### IQR

Training fences Q1 − 1.5 IQR and Q3 + 1.5 IQR. Flag strictly outside; zero-IQR columns contribute zero.

### GAUSSIAN REFERENCE ONLY

Per-column tails: 0.27% beyond 3σ, 0.70% beyond the IQR fences (about 2.7σ). Fitted bounds and multiple columns change row-level rates.

### GREAT EXPECTATIONS

Not-null on all columns; inclusive numeric training quantiles 0.1% / 99.9%. Omit undefined or equal bounds. Score: fraction of active checks violated.

8 SIX NEURAL MODELS, OPT-IN

# Neural detectors behind the same contract

PyOD 3.6.7 with recorded adaptations, CPU, seeded, fitted on at most 20,000 rows. Frozen representation scaling and latent-mean scoring change the evaluated algorithms; they do not establish equivalence to upstream defaults.

| Detector | Principle | Fixed or noted |
|---|---|---|
| Autoencoder | rebuild standardized rows through a bottleneck | MLP from scikit-learn |
| VAE | Gaussian code, rebuild, reconstruction error | scored with the code mean, PyOD draws one |
| Deep SVDD | map normal rows near a centre, distance | weakest on most types |
| DIF | isolation forests on random neural features | training representation scaler reused |
| AE1SVM | autoencoder + random-Fourier one-class SVM | singleton padding changes training weight |
| LUNAR | network learns the score from neighbour distances | best on drift, many false alarms |

The LSTM autoencoder reads windows of ordered rows and stays outside the registry.

8A EVALUATED VARIANTS

# Stable row scores require explicit adaptations

A fitted ETL detector must score a row without consulting other evaluation rows. Contract tests check the 17 row-level detectors; the time-window monitor has a different contract.

### ECOD AND COPOD

PyOD 3.6.7 recomputes tails and skew with the evaluated batch. Here ranks and skew signs are frozen from training. Even singleton scores can differ when the skew sign changes.

### COPULA MOTIVATION VERSUS IMPLEMENTED SCORE

COPOD adds marginal contributions: max(skew-selected log tail, mean of left/right log tails). This is not a fitted general joint copula. ECOD takes the max of both log tails and the skew term.

### NEURAL SCORING

VAE decodes the latent mean; DIF reuses a training representation scaler. These are algorithm changes. AE1SVM training padding also reweights its last row.

### EVIDENCE BOUNDARY

Bounded tests support batch independence, not universal equivalence. Separate ablations remain open; historical results retain their actual adapted configurations.

8B EXPLORATORY CORRECTION

# Historical pool preparation needs revalidation

Column filtering, category vocabulary and conflicting-label removal occur before the historical split. Training-only medians and scaling do not isolate those earlier decisions.

### CONSTRUCTED NEGATIVE CONTROLS

Changing only evaluation categories changes encoded width from 5 to 3. An evaluation duplicate's label can also remove a reference row. AP bias is not yet quantified.

### VERSIONED PATH

pool-train-only-v1 consumes source-bound frozen positions. Fitted schema, imputation and scaling use training rows only. Cross-partition duplicates are rejected; evaluation rows are retained.

### BOUNDED CHECKS

Separate constructed checks enforce full line/branch coverage for the two new modules and detect three leakage defects. Targets are excluded from preparation; only training prevalence calibrates flags.

### STILL OUTSTANDING

Historical memberships were recomputed (not originally archived) and replay-validated: the Z-score reproduces all 100 archived per-seed results. Replaying the studies on the train-only path remains outstanding; old protocols and results are preserved and no corrected pool-performance claim is available.

# Three example datasets bracket the method

Stratified 70 / 15 / 15 split, fit on train, every number reported on the held-out test set. Five seeds, mean ± std. The ML contamination is read from the train-set prevalence, not fixed at 10%.

**0.01 → 0.20**

Isolation Forest fraud precision, from the contamination change alone

| Dataset | Rows | Feat. | Anom. | Role |
|---|---|---|---|---|
| Credit Card Fraud | 283,726 | 30 | 0.17% | rare, but a mixed-defect label |
| Cardio | 1,831 | 21 | 9.6% | rare and outlying, the fitting regime |
| Water Quality | 3,276 | 9 | 61% | target class common, wrong regime |
| Synthetic per-type | 12,360 | 9 | 17.5% | controlled, 6 labelled types |

LOF and One-Class SVM run everywhere except Credit Card Fraud, where neither scales to 284k rows in the pipeline.

10 HOW EACH TYPE IS INJECTED

# Magnitudes are a swept parameter, not a fixed choice

**severity = 1.0** is the reference dataset. Missing values, duplicates and invalid formats have no magnitude to scale. The severity sweep (slide 15) turns "we picked the magnitudes" into a curve and shows the per-type ranking is stable. Five types replace clean rows; duplicates are appended.

| | |
|---|---|
| **MISSING VALUE** | 1–2 of {income, balance, avg amount, tenure, n txn, email, country} → null |
| **DUPLICATE** | exact copy of a sampled clean row, id included |
| **UNIVARIATE** | one numeric column × U(15, 45) × severity |
| **MULTIVARIATE** | marginals at the (1 − 0.03 / severity) percentile of the clean data |
| **TEMPORAL DRIFT** | signup in the last 80 days; income × (1 + 0.55 · severity), similar for balance and amount |
| **INVALID FORMAT** | email → malformed; age → impossible; country → out of domain |

At `severity = 1` these are the reference magnitudes. Sweeping the multiplier (slide 15) shows every distributional detector's recall on univariate, multivariate and drift rises smoothly, and the ranking does not change.

*Aggregate scores do not establish equivalence.* Per-type results show different competencies.



F1 spans 0.22 to 0.59 over the seventeen detectors on the synthetic set · the per-type competences barely overlap

# The operating point decides the outcome

The statistical rules recover most frauds at precision below 0.04. Isolation Forest changes once its contamination matches the prevalence: it flags 0.17% of rows and reaches the first usable precision, at recall 0.21 and with real seed variance (F1 = 0.20 ± 0.07). Great Expectations leads AP.

**0.20**
Isolation Forest precision (was 0.01 at contamination 0.10)

**0.182**
Great Expectations AP, the best on this dataset

| Detector | Prec | Rec | F1 | ROC | PR | Flag% | Time |
|---|---|---|---|---|---|---|---|
| Z-score | 0.011 | 0.870 | 0.021 | 0.926 | 0.078 | 13.4 | 0.03s |
| IQR | 0.003 | **0.966** | 0.007 | 0.911 | 0.018 | 48.6 | 0.12s |
| Great Expectations | 0.037 | 0.676 | 0.070 | 0.830 | **0.182** | 3.1 | 1.37s |
| Isolation Forest | **0.202** | 0.206 | **0.203** | **0.934** | 0.120 | 0.2 | 1.78s |

Mean over 5 seeds, 15% held-out test split. LOF and One-Class SVM do not scale to 284k rows here.

# When anomalies are rare and outlying, the learned detectors work

Isolation Forest reaches F1 0.62 and AP 0.70, with ECOD, COPOD and DIF close behind (AP 0.69, 0.69, 0.67). Z-score also does well: the 21 features are standardized and anomalies stand out on several at once. This is the counterpoint to a 0.17% mixed label and a 61% majority.

**0.62**
Isolation Forest F1 on Cardio

**0.70**
Isolation Forest AP on Cardio

| Detector | Prec | Rec | F1 | ROC | PR | Flag% |
|---|---|---|---|---|---|---|
| Z-score | 0.49 | **0.82** | 0.61 | 0.93 | 0.61 | 16.6 |
| IQR | 0.21 | 0.90 | 0.34 | 0.78 | 0.30 | 43.4 |
| Great Expectations | 0.51 | 0.21 | 0.29 | 0.59 | 0.19 | 4.6 |
| Isolation Forest | **0.64** | 0.62 | **0.62** | **0.95** | **0.70** | 9.9 |
| LOF | 0.25 | 0.27 | 0.26 | 0.62 | 0.24 | 10.7 |
| One-Class SVM | 0.56 | 0.61 | 0.58 | 0.94 | 0.63 | 10.7 |

Cardiotocography, 1,831 rows, 9.6% anomalies. Mean over 5 seeds, test split. Wide seed spread on this small set (ML F1 std up to 0.17).

# Aggregate scores hide different trade-offs

Z-score and HBOS reach a perfect precision by firing only on extreme rows (recall 0.20). DIF has the best F1 and AP, the VAE matches Isolation Forest, LUNAR has the best recall with a low precision.

**0.22–0.59**

F1 range across the seventeen detectors

| Detector | Prec | Rec | F1 | ROC | PR |
|---|---|---|---|---|---|
| RULES AND CLASSICAL | | | | | |
| Z-score | **1.000** | 0.196 | 0.328 | 0.742 | 0.544 |
| IQR | 0.584 | 0.396 | 0.472 | 0.675 | 0.452 |
| Isolation Forest | 0.495 | 0.499 | 0.497 | 0.735 | 0.485 |
| One-Class SVM | 0.478 | 0.485 | 0.481 | 0.722 | 0.520 |
| ECOD | 0.341 | 0.341 | 0.341 | 0.675 | 0.283 |
| HBOS | **1.000** | 0.204 | 0.339 | 0.602 | 0.343 |
| KNN | 0.466 | 0.469 | 0.467 | 0.745 | 0.520 |
| NEURAL | | | | | |
| VAE | 0.502 | 0.507 | 0.504 | 0.742 | 0.548 |
| DIF | 0.589 | 0.582 | **0.585** | **0.772** | **0.648** |
| LUNAR | 0.345 | **0.668** | 0.410 | 0.728 | 0.495 |

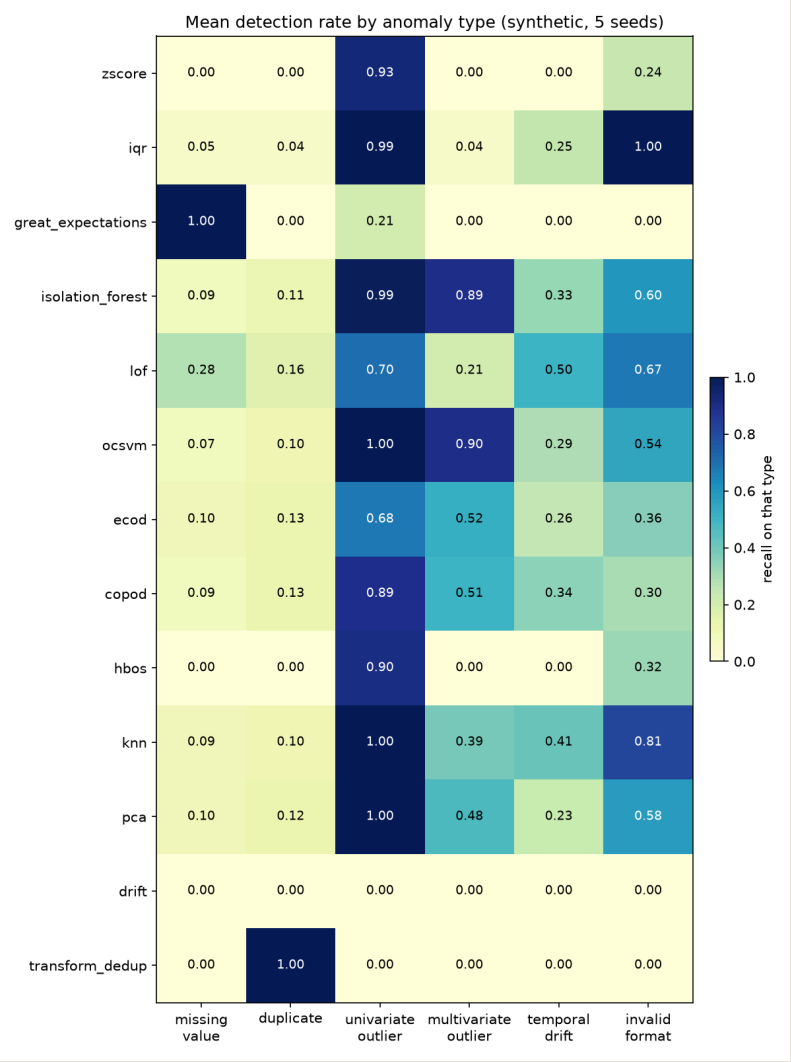
Mean over 5 seeds, test split. 12,360 rows, 17.5% anomalous. All 17 detectors are in the paper.

# Each method has a narrow competence

Z-score owns univariate outliers. Great Expectations owns missing values (recall 1.00, precision 0.90). Deduplication owns exact duplicates. Isolation Forest and One-Class SVM are the only methods that catch multivariate outliers. Broad coverage needs several families, not the single best one.

## 0.91 vs ≤0.02

Isolation Forest recall on multivariate outliers, against every rule




paper/figures/heatmap_synthetic.png

transform_dedup and drift shown for reference

16 RECALL AND PRECISION, PER TYPE

# Detection rate alone can be gamed

Isolation Forest catches 91% of multivariate outliers at only 0.24 precision: a wide net that also pulls in clean rows. Z-score fires only when it is certain, so its precision is 1.00 on the two columns it covers.

**0.91 / 0.24**

recall / precision on multivariate outliers, Isolation Forest

| Method | missing | | univ. out. | | multiv. out. | | drift | |
|---|---|---|---|---|---|---|---|---|
| | rec | prec | rec | prec | rec | prec | rec | prec |
| Z-score | 0.00 | – | **0.92** | **1.00** | 0.00 | – | 0.00 | – |
| IQR | 0.05 | 0.03 | **0.99** | 0.39 | 0.02 | 0.01 | 0.27 | 0.15 |
| Great Expectations | **1.00** | **0.90** | 0.23 | 0.70 | 0.00 | – | 0.00 | – |
| Isolation Forest | 0.09 | 0.03 | **0.99** | 0.26 | **0.91** | 0.24 | 0.33 | 0.10 |
| LOF | 0.26 | 0.06 | 0.65 | 0.15 | 0.16 | 0.04 | **0.49** | 0.11 |
| One-Class SVM | 0.08 | 0.03 | **0.99** | 0.25 | 0.84 | 0.22 | 0.34 | 0.10 |
| transform dedup | 0.00 | – | 0.00 | – | 0.00 | – | 0.00 | – |

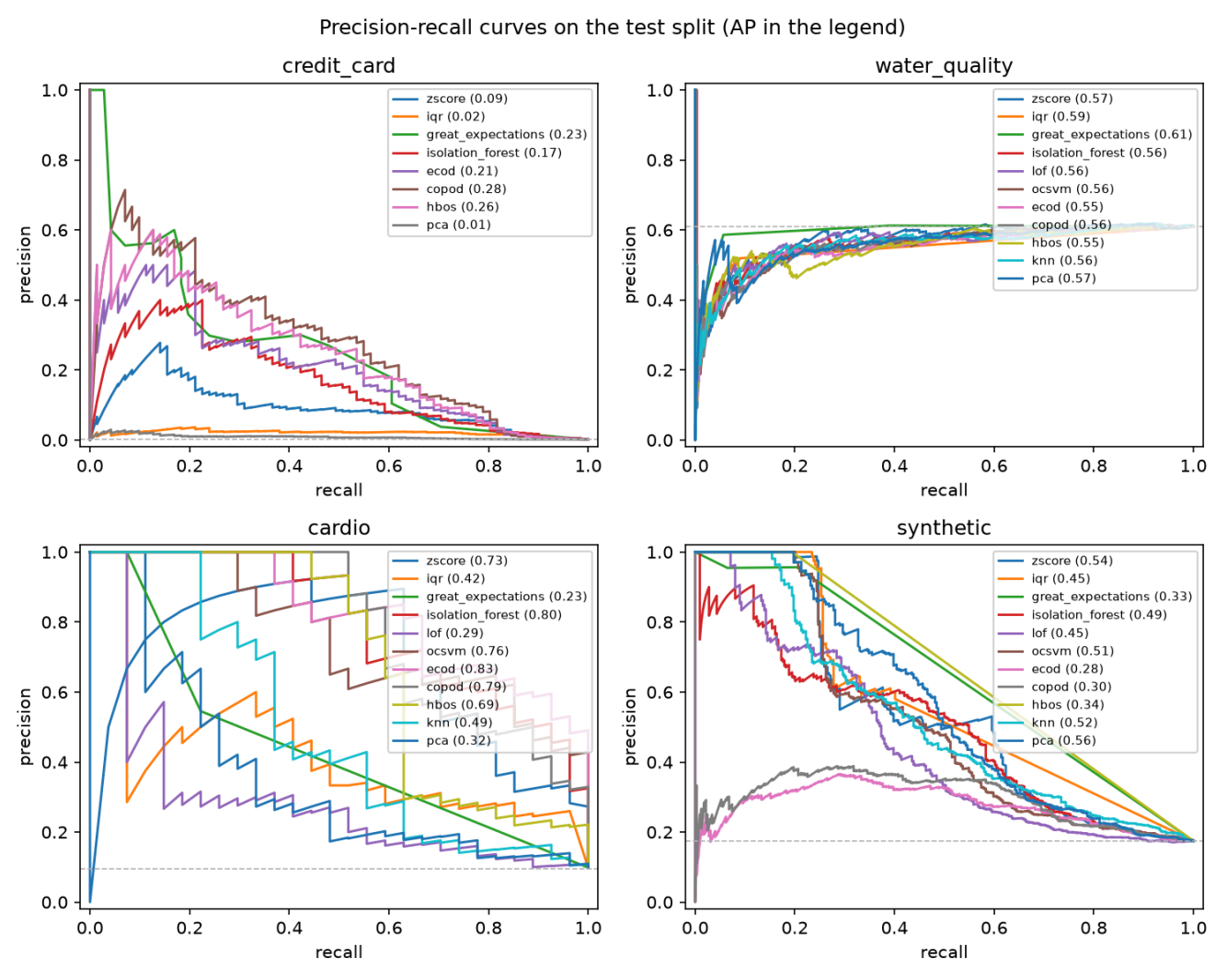
Recall and precision per anomaly type, clean rows as the only negatives. 5 seeds, mean. Duplicate and invalid-format columns omitted for space.

# One threshold hides the whole trade-off

On Cardio the Isolation Forest and One-Class SVM curves stay high (AP 0.80, 0.76). On Water Quality every curve hugs the 0.61 base-rate line. On Credit Card Fraud two methods dominate but neither clears precision 0.4 except at negligible recall.

## 0.37 → 0.008

Isolation Forest fraud precision as recall goes 0.23 → 0.92

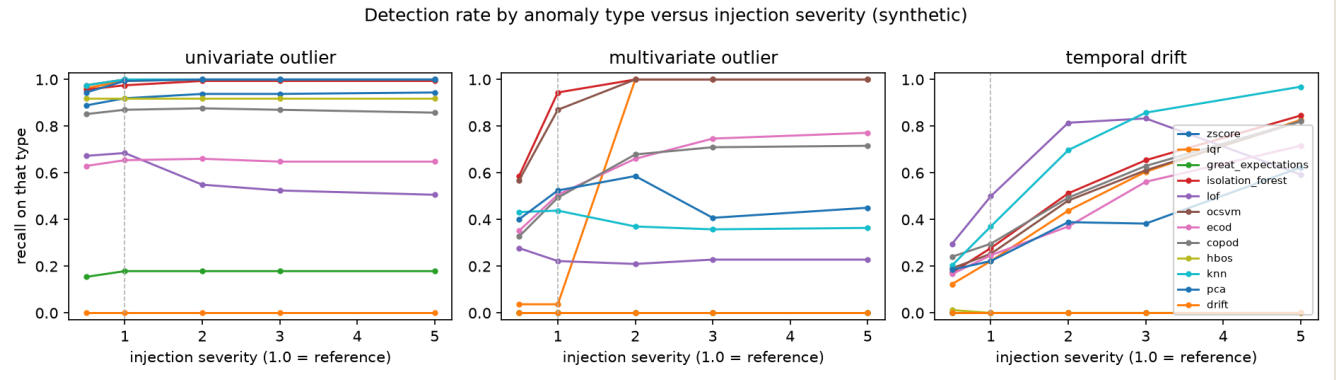


# The per-type ranking is stable as the magnitudes scale

Univariate outliers are saturated already at severity 0.5. Multivariate outliers show a threshold. Temporal drift climbs roughly linearly for the distributional detectors: the reference drift is simply below every method's floor, not a property of the methods. The same families lead the same columns at every severity.

**≤0.3 → 0.85**

temporal-drift recall from severity 1 to 3



# A pool of 152 open datasets, pinned and checked

Issue #66. The three datasets of M1 bracket the method but are too few to say anything general. The pool follows ADBench in scale and adds a licence rule: a dataset enters only when its terms are read and open. Raw data stays out of the repository; checksums, licences and reports are archived.

**152**
pinned datasets

**116**
clean carriers for injection

**100**
labelled anomaly problems

| | |
|---|---|
| ADBench tabular sets kept (35 CC BY 4.0, 1 ODbL) | 36 OF 47 |
| UCI sets passing metadata and licence filters | 133 |
| UCI sets kept after curation, reason recorded | 116 |
| anomaly label rule fixed before any detector | RAREST CLASS |
| one normalisation policy for all sets | SINGLE |
| download check of every set, manual and monthly CI | PINNED |
| excluded for unclear terms (celeba, mammography, vowels...) | 11 |

# Historical pool rankings need train-only revalidation

Historical run: three rules, eight classical and six neural detectors, five seeds, on 36 ADBench and 64 UCI problems. The numerical table is retained as recorded. Schema and row filtering precede the split, so these descriptive rankings still need train-only replay; the 100 problems come from 80 sources and a source-clustered re-analysis changes no sign (only Great Expectations is clearly below Isolation Forest); label-rule sensitivity and unrecorded PyOD convergence diagnostics remain open.

**17**
detectors; no neural exception recorded

**13**
problems won by DIF, the most; each of the 17 wins at least 2

| Detector | Problems | AP median [95%] | ROC-AUC | Rank |
|---|---|---|---|---|
| ECOD | 100 | **0.240 [0.191, 0.305]** | 0.656 | 7.0 |
| One-Class SVM | 84 | 0.240 [0.177, 0.307] | 0.670 | 8.0 |
| HBOS | 100 | 0.237 [0.176, 0.311] | 0.663 | 8.0 |
| DIF | 100 | 0.226 [0.173, 0.272] | 0.639 | 8.0 |
| VAE | 100 | 0.216 [0.163, 0.279] | 0.676 | 7.0 |
| COPOD | 100 | 0.216 [0.162, 0.314] | **0.691** | 7.0 |
| Isolation Forest | 100 | 0.215 [0.158, 0.293] | 0.677 | 7.0 |
| AE1SVM | 100 | 0.210 [0.172, 0.253] | 0.635 | 8.0 |
| Autoencoder | 100 | 0.205 [0.150, 0.240] | 0.631 | 8.0 |
| Z-score | 100 | 0.203 [0.164, 0.279] | 0.626 | 9.0 |
| KNN | 92 | 0.203 [0.172, 0.284] | 0.658 | 8.0 |
| LUNAR | 100 | 0.183 [0.153, 0.225] | 0.615 | 10.0 |
| Deep SVDD | 100 | 0.181 [0.147, 0.226] | 0.619 | 10.0 |
| PCA | 100 | 0.178 [0.152, 0.219] | 0.581 | 10.0 |
| LOF | 92 | 0.165 [0.153, 0.237] | 0.588 | 10.0 |
| IQR | 100 | 0.144 [0.117, 0.191] | 0.556 | 12.0 |
| Great Expectations | 100 | 0.115 [0.105, 0.123] | 0.508 | 16.0 |

Median over problems, sorted by AP. LOF and KNN are skipped above 50,000 training rows and One-Class SVM above 20,000. Neural models fitted on at most 20,000 rows.



labels from a rare class, not from data-quality defects

# Two regimes the three examples do not show

Among the 100 problems, the largest margin of the rules over the learned models is on Annthyroid (IQR 0.61, best learned 0.31). The largest margin of a neural model is on Mushroom (LUNAR 0.96, best classical 0.63; label derived by a rule). Chosen after the results, as illustrations, not as a test.

**0.61**
IQR rule on Annthyroid (best learned model 0.31)

**0.96**
LUNAR on Mushroom, where the IQR rule is at chance (0.10)

| AP | Credit Card | Cardio | Annthyroid | Mushroom |
|---|---|---|---|---|
| Z-score | 0.10 | 0.61 | 0.18 | 0.44 |
| IQR | 0.02 | 0.30 | **0.61** | 0.10 |
| Isolation Forest | 0.13 | **0.70** | 0.29 | 0.63 |
| ECOD | 0.21 | 0.69 | 0.28 | 0.41 |
| HBOS | 0.26 | 0.58 | 0.24 | 0.50 |
| Autoencoder | 0.11 | 0.35 | 0.14 | 0.68 |
| DIF | 0.27 | 0.67 | 0.23 | 0.45 |
| LUNAR | 0.15 | 0.33 | 0.20 | **0.96** |

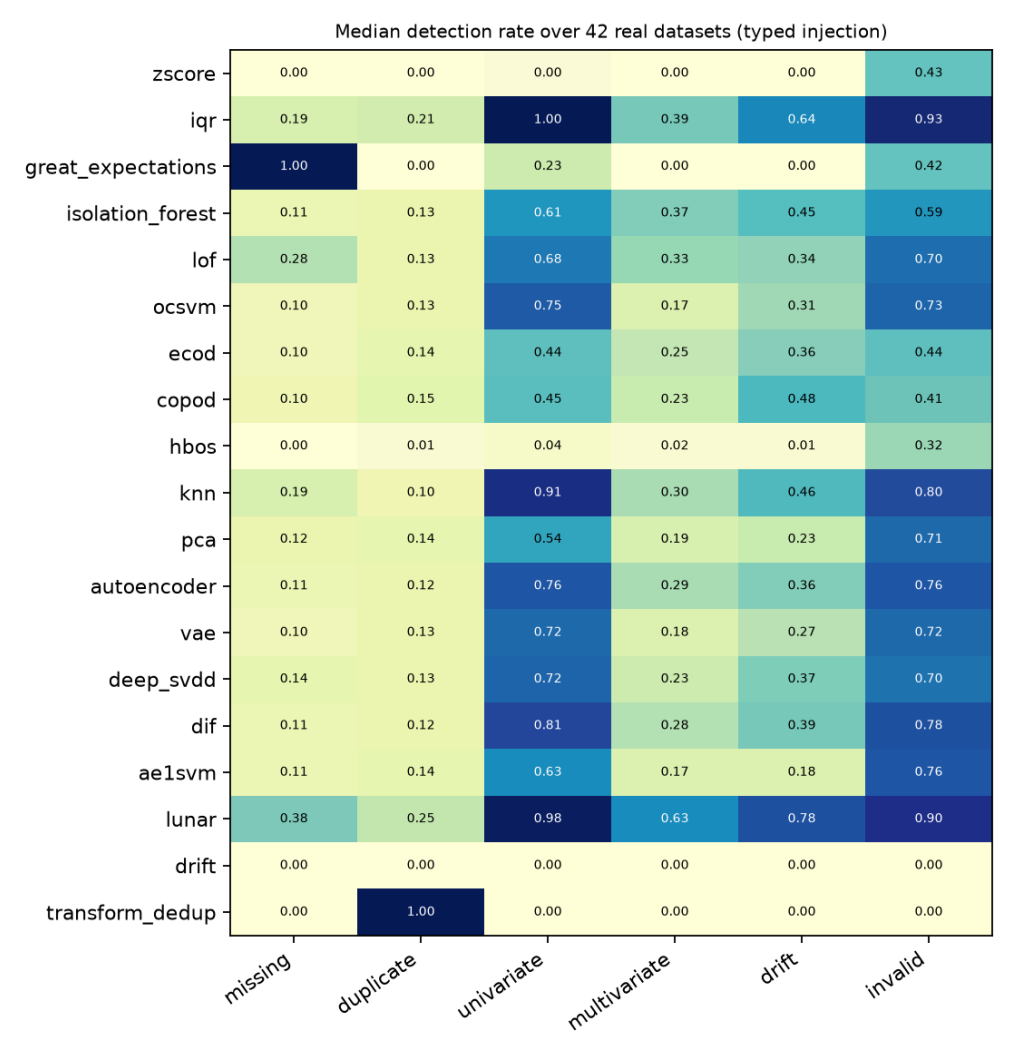
Mean over 5 seeds, as run in the pool (same normalisation for all). Best of each column in green; all 17 detectors are in the paper.

# The synthetic pattern does not simply carry over

Issue #65. The six anomaly types are injected into 42 real UCI tables (3% of rows per type, same split and settings as the synthetic run). Protocol and five hypotheses were written before any run. Two rows hold everywhere: Great Expectations finds all missing values, deduplication all duplicates. Three hypotheses are not supported. The eleven detectors added later run on the same tables: LUNAR is best on multivariate outliers (0.63) and drift (0.78), but flags a median 24% of clean rows.

**0.37**
Isolation Forest, multivariate outliers (0.89 synthetic)

**24%**
median clean rows flagged by LUNAR (IQR 22%)






median over 42 UCI tables (58 with ADBench, same picture), injected not natural

# Two failures that are explained, not hidden

Z-score detects almost nothing on univariate outliers (median 0.00 against 0.93 synthetic). A check made after seeing it, so exploratory, shows why: invalid rows in the training split inflate the standard deviation. Without them the Z-score finds every univariate outlier on all 42 tables. The IQR fences reach the best detection rate on several types, but only by flagging a median 22% of clean rows.

**0.83**
mean rank correlation between two tables

**-0.96**
IF detection vs number of columns

### MASKING

Non-robust statistics on a contaminated training set, not a weakness of the Z-score as such.

### DETECTION RATE CAN BE GAMED

The false positive rate on clean rows has to be read next to it. It reaches 99% on some wide tables.

### STABLE RANKING

Rank correlation of the method-by-type matrices is 0.83, lowest for drift (0.70). Partly mechanical: many cells sit near 0 or 1.

### DIMENSION HURTS

Isolation Forest on univariate outliers falls as the number of continuous columns grows.

# Time changes the answer: a threshold from training does not travel

Credit Card is the only dataset with a clock. Detectors are fitted on the first 70% of time and scored on the last 15% (52 frauds). Average precision drops for all four detectors, but the interval excludes the stratified mean only for Great Expectations and Isolation Forest. Prevalence also changes between partitions, so the pilot does not separate it from the effect of time.

**0.936**
Isolation Forest ROC-AUC, chronological

**0.08%**
future rows it flags: none of the 52 frauds

| Detector | PR strat. | PR chrono | ROC chrono | Flag% |
|---|---|---|---|---|
| Z-score | 0.078 | 0.048 | 0.935 | 12.8 |
| IQR | 0.018 | 0.013 | 0.917 | 47.1 |
| Great Expectations | 0.183 | 0.045 | 0.758 | 3.5 |
| Isolation Forest | 0.110 | 0.035 | 0.936 | 0.08 |

Average precision on the test split, 95% block-bootstrap interval for the chronological arm in the paper. The window monitor detects mean shifts of 1 and 2 standard deviations, not 0.5.

# Part of the gap is only the prevalence

Issue #63. Credit Card and Bank Marketing (UCI, ordered by date 2008 to 2010), four detectors, three label-free thresholds. On Bank Marketing the prevalence goes from 5.8% in training to 39.3% in test: matching the stratified control to it removes most of the gap. A rolling threshold that keeps the training prevalence cannot follow it. One hypothesis of four holds.

**0.26 → 0.06**

median AP gap on Bank Marketing after prevalence matching

| F1 at the threshold | P0 training | P2 rolling |
|---|---|---|
| Bank, Z-score | 0.51 | 0.02 |
| Bank, Isolation Forest | 0.41 | 0.11 |
| Credit Card, Z-score | 0.00 | 0.09 |
| Credit Card, Great Expectations | 0.00 | 0.09 |

Test period, threshold at the training prevalence. Rank correlation of the detector order, stratified against chronological: 0.80 on Bank Marketing, 0.40 on Credit Card. Kyoto-2006+ left out: rights and download not verified.

25A M2 · EXPLORATORY DEPLOYMENT THRESHOLDS

# A nominal alert budget is not a promise

Issue #94, exploratory. Budgets of 0.5 to 10% set from training scores only, on 100 labelled problems. Held-out alert volume was 1.05 to 1.36 times the nominal 1% budget for classical detectors and 1.13 to 2.87 times for neural ones. One machine; not in the main tables.

**+4.1×**

alert volume, in budgets, for a 1 σ score shift; ROC-AUC unchanged

| Change at a 2% budget | Δ alert ÷ budget [95%] | Δ ROC-AUC |
|---|---|---|
| Evaluated scores shifted +1 σ | +4.09 [3.50, 4.77] | 0.000 |
| Evaluated scores scaled ×2 | +4.86 [4.57, 5.19] | 0.000 |
| Noise of 1 σ on the scores | +1.55 [1.32, 1.81] | −0.038 |
| Every feature drifted ½ σ | +20.90 [17.39, 24.55] | −0.051 |
| Monotone change of fit and test scores | +0.001 [0.000, 0.003] | 0.000 |

Five classical detectors, 98 problems from 78 sources, sources resampled whole, Holm-adjusted. Prevalence reweighted under a label-shift assumption: 0.83 to 1.83 times the budget at 0.5% prevalence, 1.15 to 2.73 times at 20%. The 30-expected-alert rule for monitoring was chosen after the false alarms were seen.

# The deeper studies, run on all 152 datasets

Each study ran on every dataset where it is defined; the others are listed with a reason. Six hypotheses were written first. Three hold and three do not, and the failures are reported as such.

**58 / 152**
tables eligible for typed injection and severity

**290 / 304**
end-to-end pipeline runs that completed

| Hypothesis | Result | Value |
|---|---|---|
| P1 label-free threshold lowers F1 | **holds** | 14 of 17 detectors |
| P2 curve ranking = AP ranking | fails | Spearman 0.90 (< 0.99) |
| P3 recall rises with severity | fails | 13 of 17; column-wise rules stay at 0 |
| P4 detection share of the pipeline | **holds** | 62% (IF), 4% (Z-score), 14 datasets |
| P5 chronological AP lower | fails | 31 of 64 pairs; Bank Marketing 0/16 |
| P6 files match the manifest | **holds** | 152 of 152 |

Chronology needs a real time variable: 5 labelled datasets qualify (ADBench has none).

# An LSTM autoencoder is comparable, not clearly better

Issue #52. The window is the ten previous rows of the global stream, since Credit Card has no cardholder. The protocol and a CPU budget of 20 epochs were fixed beforehand. On a synthetic stream with broken continuity the LSTM wins by a wide margin. On the real frauds it only matches Isolation Forest and Great Expectations, and none of five trainings converged within the budget.

**0.875**
LSTM AP on the synthetic stream (row AE 0.006)

**0.037**
LSTM AP on Credit Card (row AE 0.024)

| | |
|---|---|
| paired bootstrap, main seed | +0.021 [+0.005, +0.044] |
| other four seeds | INTERVAL INCLUDES 0 |
| versus Isolation Forest (0.035), GX (0.045) | COMPARABLE |
| convergence within 20 epochs | NOT REACHED |
| recall of frauds in the later period | ≤ 0.02 |
| deep models in the per-type matrix | #64 DONE |

# The profile is stable, two cells are not

Issue #67. Transactions and sensor readings get the same six injected types and the same protocol as the customer table. Great Expectations and the deduplication rule carry over exactly, and the rank correlation between schemas is 0.93. The multivariate column and drift depend on how the columns are scaled and correlated. Two of four hypotheses hold.

**0.93**
mean rank correlation between schemas

**0.96**
best drift recall on sensors (0.38 on transactions)

| Multivariate outliers | customers | transactions | sensors |
|---|---|---|---|
| Isolation Forest | 0.89 | 0.49 | 0.32 |
| One-Class SVM | 0.90 | 0.18 | 0.33 |
| IQR | 0.04 | 0.57 | 0.02 |
| Z-score | 0.00 | 0.00 | 0.00 |

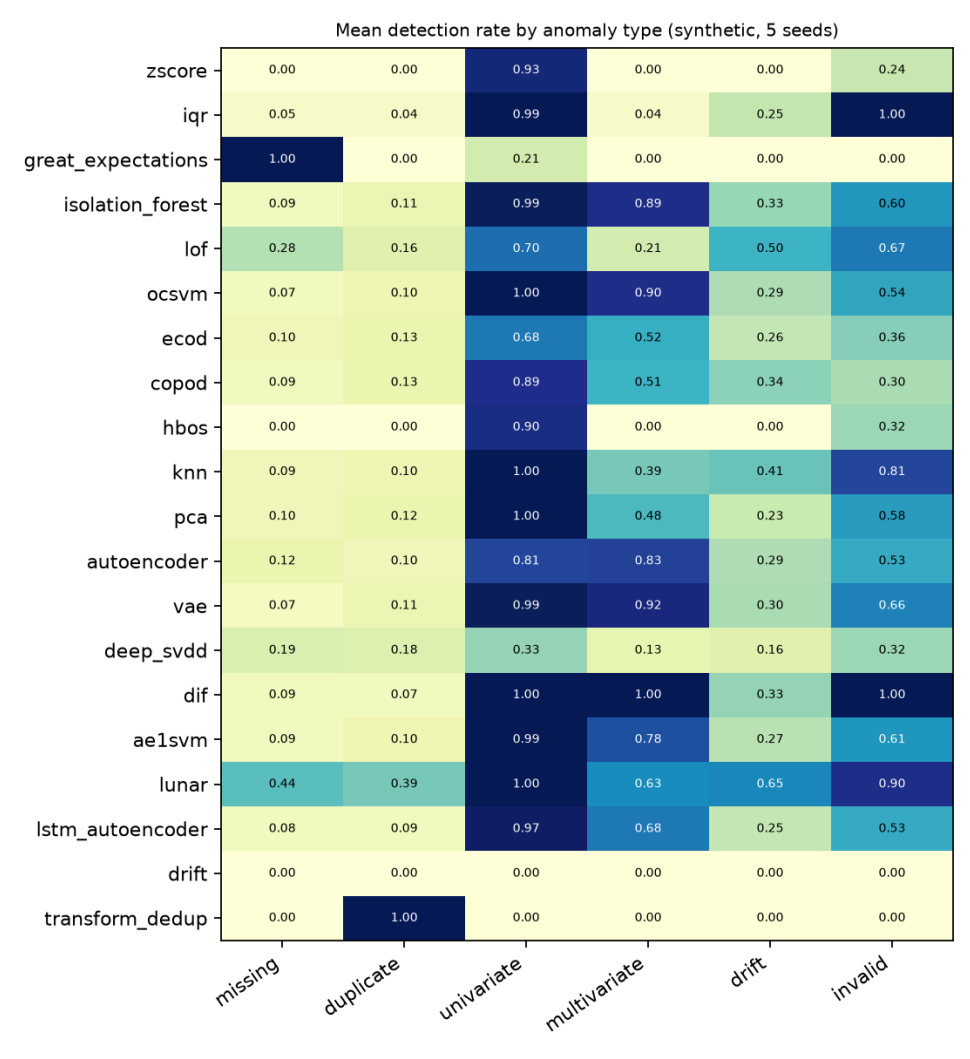
Mean recall over five seeds. Same generator logic for all three schemas: the agreement shows the profile does not depend on one table, not that it holds on real data.

# More detectors, same competence map, one exception

Eleven rows added to the synthetic matrix, with three hypotheses written beforehand; none holds. ECOD and COPOD find 0.5 of the multivariate outliers although they read one column at a time, PCA reaches 0.48 against 0.83 for the autoencoder, and LUNAR catches 0.65 of the drifted rows (LOF 0.50) while flagging 41% of clean rows. The LSTM still does not fix drift.

**0.65**
LUNAR recall on drift, with 41% of clean rows flagged

**1.00**
DIF recall on multivariate outliers



# Detection latency is not pipeline latency

Issue #53. Synthetic matrices from 1,000 to 300,000 rows, a fresh process per measurement so that peak memory belongs to that run, 300 s budget. On the full Credit Card file the Z-score pipeline takes 4.8 s of which detection is 3%, while Isolation Forest takes 8.5 s with detection at 43 to 45%.

**133 s**
LOF at 300,000 rows (exponent 1.51)

**>300 s**
One-Class SVM at 100,000 rows

| Detector | 10k rows | 100k rows | 300k rows | Memory MB |
|---|---|---|---|---|
| Z-score | 0.002 s | 0.013 s | 0.039 s | 70 |
| IQR | 0.004 s | 0.029 s | 0.086 s | 116 |
| Isolation Forest | 0.16 s | 1.09 s | 3.2 s | 140 |
| LOF | 0.73 s | 18.0 s | 133 s | 405 |
| One-Class SVM | 5.6 s | >300 s | not run | 788 |
| ECOD / COPOD | 0.045 s | 0.68 s | 2.4 s | 186 |
| KNN | 0.45 s | 11 s | 68 s | 148 |
| VAE | 6.8 s | 69 s | 209 s | 105 |
| DIF | 23 s | 262 s | >300 s | 2,748 |

Median detection-stage time, 8 features; five of the ten newer detectors shown (paper has all). Autoencoders and the other neural models are in the paper.

31 THE PRACTICAL OUTPUT

# Anomaly type to method

Two of six types are not a per-row detector problem at all: duplicates belong to the transform stage, drift to a windowed monitor. That itself is a finding a single aggregate ranking would never surface.

| Anomaly type | Use | Recall | Why |
|---|---|---|---|
| missing value | Great Expectations | 1.00 | not-null expectation, exact |
| exact duplicate | transform dedup stage | 1.00 | row equality, not a detector's job |
| univariate outlier | any statistical rule | 0.92–0.99 | single-column band is enough |
| multivariate outlier | Isolation Forest | 0.91 | only the joint models see it (precision 0.24) |
| temporal drift | windowed monitor | ≤0.49 | sub-threshold per row; needs a stage |
| invalid format | IQR, partially | 1.00 / none | numeric part caught, string format missed |

# Unsupervised detection works in one of three regimes

The question is less "rules or ML" than "does this dataset have rare, outlying anomalies at all". Latency and scaling then constrain which detector can sit in a real flow: Great Expectations is the slowest linear method, and the One-Class SVM and LOF do not survive the full fraud dataset.

## Fraud

### too extreme

0.17% positives, and the label mixes every defect. Best F1 of seventeen detectors is 0.35 (Deep SVDD), best precision 0.34 (HBOS).

## Cardio

### the fit

9.6%, rare and distinct. Isolation Forest F1 0.62, AP 0.70. The regime the detectors are built for.

## Water

### wrong regime

61% majority target. Every ROC-AUC 0.45 to 0.51: the detectors are at chance.

The synthetic dataset sits in the middle at 17.5% and is where the per-type structure is measured. Pipeline cost narrows the shortlist further: Great Expectations is the slowest linear method, and the One-Class SVM and LOF do not survive the full fraud dataset.

33 WHAT WAS HARDER THAN THE PLAN ASSUMED

# Findings from the build

Each of these became a paragraph in the paper. The recurring lesson: the setup around a detector, the operating point, the regime, the pipeline cost, matters as much as the algorithm.

### THE OPERATING POINT IS A LEVER

Contamination = prevalence instead of a fixed 10% moves Isolation Forest fraud precision from 0.01 to 0.20 and multivariate recall from 0.41 to 0.91.

### ONE REGIME OUT OF THREE

Cardio: F1 0.62. Fraud (mixed label): nothing usable. Water Quality (61% majority): at chance.

### GX 1.X IS A FULL API REWRITE

Per-row failures need the COMPLETE result format plus an explicit index column. Figured out by probing the installed library.

### PYOD SCORES CAN DEPEND ON THE BATCH

ECOD, COPOD and DIF use the scored rows for their statistics, the VAE draws a random code, AE1SVM crashes on a last batch of one row. Recorded adaptations, with 17 row-level batch-independence checks; not upstream-equivalence claims.

### ML IS WEAK ON MISSING VALUES

Imputing removes the signal a not-null rule catches for free: GX recall 1.00 against 0.08 to 0.26 for the learned models.

### DRIFT IS NOT A PER-ROW PROBLEM

Once diluted by clean recent rows the window mean has moved by about a tenth of a standard deviation.

# Historical verification has a defined scope

These measurements belong to local run d4304c5. They cover the core pipeline and experiment scripts, with separate verifier checks. They do not certify the new research modules or establish scientific validity. The run is not independently attested.

**1,292**
tests, plus 3 on full datasets

**97.61%**
mutants killed (4,980 of 5,102)

| Check | Result | Scope |
|---|---|---|
| line coverage | 1,790 / 1,790 | pipeline modules and experiment scripts |
| branch coverage | 350 / 350 | same scope |
| mutation, computing code | 95.68% | split, metrics, generator, detectors |
| mutation, input / output | 98.87% | readers, writers, scripts |
| fault injection | 23 / 23 | four went through every earlier test |
| real-data check | pass | three CSV files, manuscript tables match |

`Survivors (122) count as failures and were classified by reading the changes, not by proof.`

# A surviving mutant is a behaviour no test protects

## 1 CHANGE ONE THING

```
- flags = scores > info["threshold"]
+ flags = scores >= info["threshold"]
```

A real mutant of `research/alert_budget.py`. A score equal to the training threshold would now raise an alert.

## 2 RUN THE TESTS

| | |
|---|---|
| `killed` | a test fails: the change is noticed |
| `survived` | every test passes: the behaviour is not checked |

Here the test `a score equal to the training threshold is not an alert` kills it. It did not exist before this session.

## 3 SCORE THE MODULE

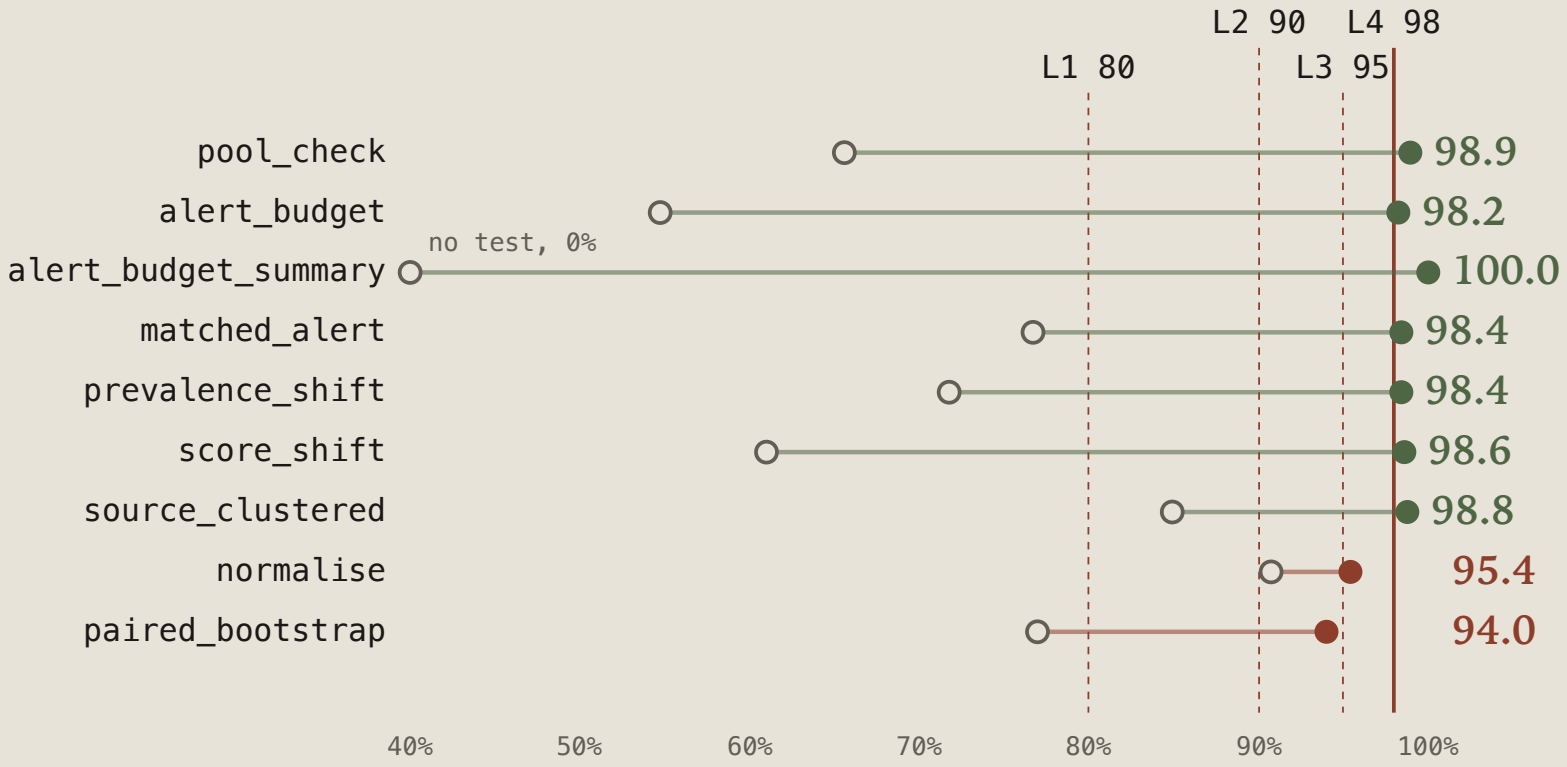
**killed**

---

**all mutants**

Survivors stay in the denominator. A mutant counts as equivalent only when the owner has recorded it as such, and never by assumption.

# Four thresholds by scientific risk, nine modules moved




Raw focused mutation score per module, local run, 2026-11-11. Open circle: before this session. Filled: after new behavioural tests. Green: at or above the L4 threshold of 98%, on that module alone. All nine modules are L4. Thresholds were not changed.

# A focused pass does not certify the repository

### LSTM AUTOENCODER, ONE MODULE, ONE SHARD

**97.61%**
raw, 368 of 377

**98.66%**
adjusted, 368 of 373

Adjusted means four equivalences accepted by the repository owner, as sole author and reviewer, not an independent review. Five inconclusive mutants stay eligible. No timeout.

### FIRST COMPLETE NATIVE CAMPAIGN, 2026-11-07, L4 MODULES




65.5% of L4 mutants killed against 98% required, 7,501 further mutants with no covering test. The release gate fails and stays closed.

**OPEN**

32 modules below threshold and 30 incomplete in that campaign, not re-measured here

**OPEN**

survivors that look inert (dtype coercions, `zip(strict=)`) await an owner decision

**FOUND**

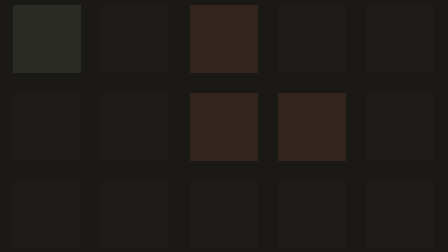
minimum anomaly count 3 is below the 6 a stratified split needs; no archived result changes

# Experiments run, paper in IMRaD form

Historical implementation issues #63 to #70 are closed. The current roadmap preserves T001–T415. 91 of the 415 tasks are checked as of 2026-11-11, including the exploratory deployment-threshold tasks T163 to T171 of #94; T172, T062 and T063 remain open, as do the issues themselves. Source-aware inference, partition recovery, replay and publication corrections remain required. Optional P2 research is outside the active scope.

| | |
|---|---|
| held-out split, 5 seeds, chronological and LSTM pilots, cost study | DONE |
| dataset pool of 152 sets (#66), typed injection on 42 tables (#65) | DONE |
| benchmark on the 100 labelled pool problems | DONE |
| deep rows (#64), multi-schema generator (#67) | DONE |
| historical IMRaD restructuring (#68, #69, #85); final bibliography audit remains #101 | DONE |
| decision-level study under temporal shift (#63) | DONE |
| paper layout (tables) and discussion revised, merged to main | DONE |
| complete registry: 17 detectors, every study rerun | DONE |
| every study extended to the 152-dataset pool, six hypotheses | DONE |
| P0 remediation: detector/source audits, source-aware statistics, archive replay, publication | NEXT |

# Rules and machine learning are complementary by anomaly type.



The per-type heatmap is what makes that choice actionable. Across the three example datasets, unsupervised detection pays off only on Cardio, where the anomalies are genuinely rare outliers. On 100 labelled problems of the pool, rankings require train-only revalidation. On 42 real tables with injected types, the synthetic pattern holds only in part, which is why the real-data matrix is the differentiator. The pipeline, the 152-dataset pool, the protocols and the verification layer are in place. The IMRaD write-up and the remaining studies are what remains.